

Performance Indicators Calculation Exercise

Let's assume we have trained a classifier to predict breast cancer in medical images. See below evaluation performance on a held-out test set for this classifier:

#	True label	Predicted label
1	yes	yes
2	no	no
3	no	yes
4	no	no
5	no	no

#	True label	Predicted label
6	yes	no
7	yes	yes
8	no	yes
9	no	no
10	no	no

#	True label	Predicted label
11	no	yes
12	yes	yes
13	no	no
14	no	no
15	yes	no

1. First, count the true positives, true negatives, false positives, and false negatives.
2. Draw the confusion matrix. On the x-axis put the true label, on the y-axis the predicted label.
3. Calculate error rate, accuracy, precision, recall and the F1 score. Find the formulas on the back of the page.
4. Given that this classifier is supposed to predict breast cancer in medical images, is it well suited for this task? Which performance indicator do you think is the most important in this case?

Performance measures

Error rate = simply the percentage of instances where the classifier was wrong

Accuracy = 1 - error rate

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$